

Python Course for Physicists (And everyone else!)

Chunk 3: Logic Operators and If-Statements

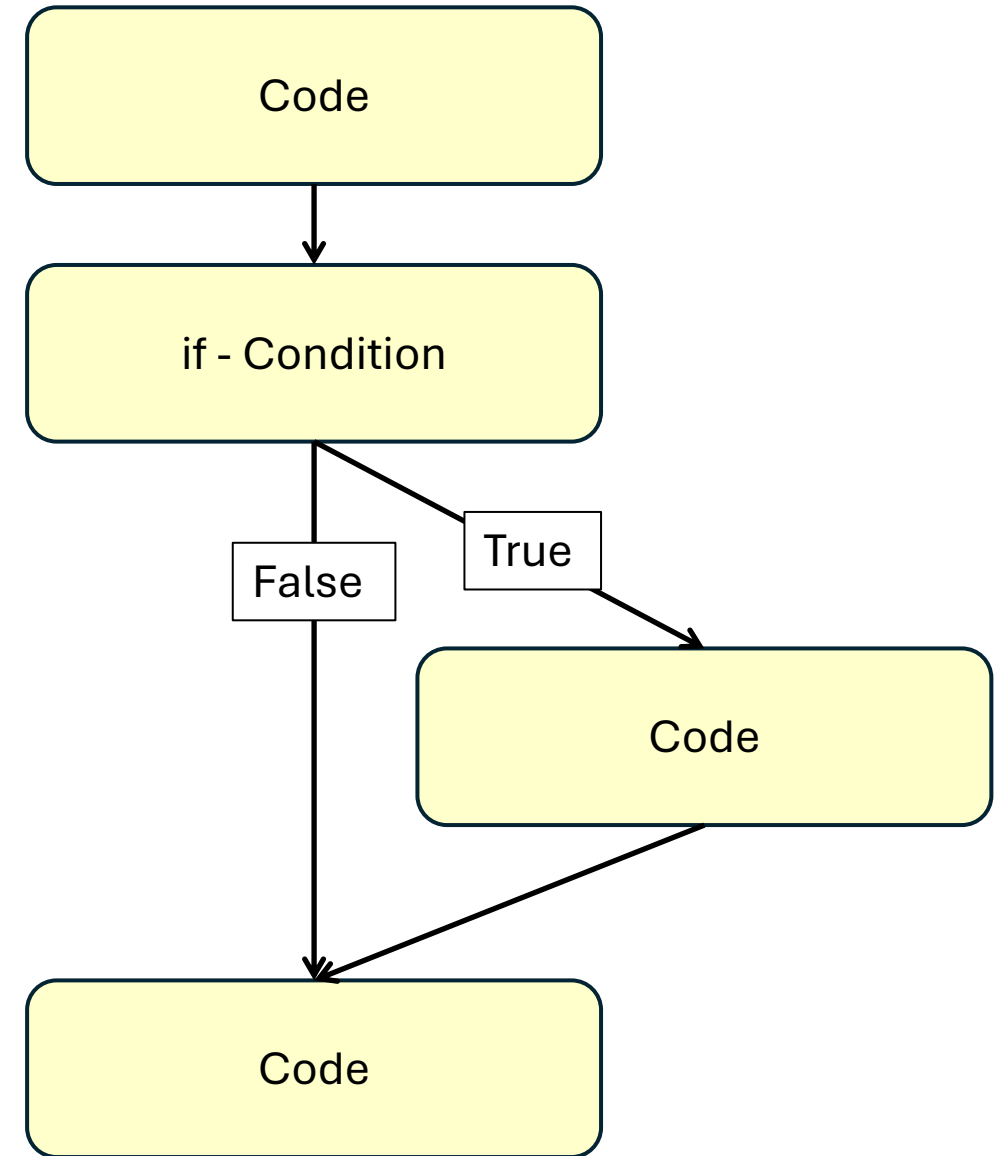
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Program Flow:

- Execute a part of code, only when Condition is true

```
# code  
if xxxxxxxx:  
    # code  
# code
```



The Condition – Comparison Operators

- We need a Boolean Condition: True or False
- Comparison Operator → True or False

```
print("Before")
if True:
    print("True")
if False:
    print("False")
print("After")
```

Before
True
After

```
a = 20 > 10
print(a)
```

True

True or False

Operator	Name	Example
==	Equal	x == y
!=	Not equal	x != y
>	Greater than	x > y
<	Less than	x < y
>=	Greater than or equal to	x >= y
<=	Less than or equal to	x <= y

Live Coding: Examples

```
name = "Emil"
if name == "Emil":
    print("The name is Emil")
if name != "Emil":
    print("The name is not Emil")
```

The name is Emil

```
name = "Dieter"
if name == "Emil":
    print("The name is Emil")
if name != "Emil":
    print("The name is not Emil")
```

The name is not Emil

```
your_age = 12
if your_age >= 18:
    print("You can buy cigarettes")
if your_age < 18:
    print("You can not buy cigarettes")
```

You can not buy cigarettes

```
your_age = 99
if your_age >= 18:
    print("You can buy cigarettes")
if your_age < 18:
    print("You can not buy cigarettes")
```

You can buy cigarettes

Anecdote: Traction Control

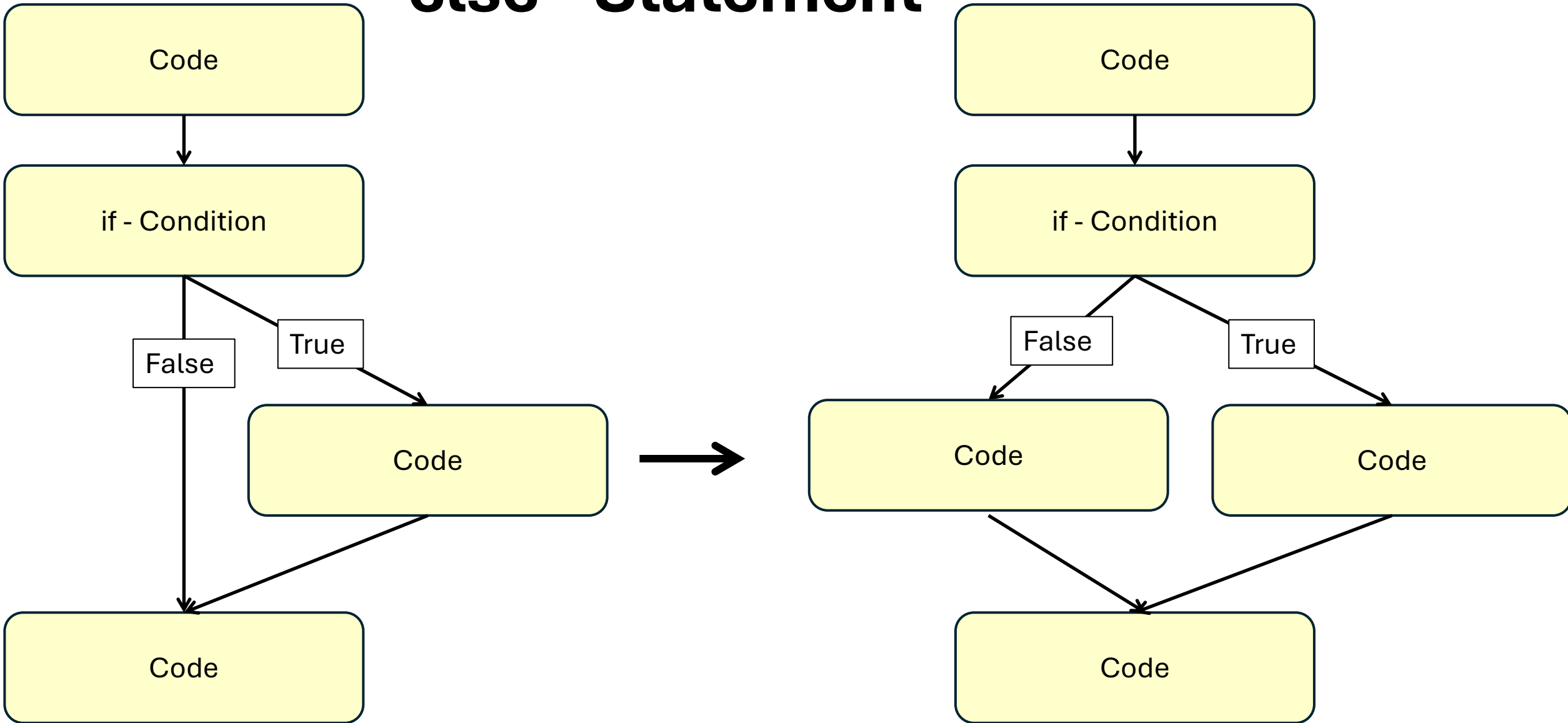
```
static void TractionControl(float& throtmin, float& throtmax)
{
    if (!Param::GetBool(Param::espoff))
    {
        float frontAxleSpeed = (Param::GetFloat(Param::wheelfl) + Param::GetFloat(Param::wheelfr)) / 2;
        float rearAxleSpeed = (Param::GetFloat(Param::wheelrl) + Param::GetFloat(Param::wheelrr)) / 2;
        float diff = frontAxleSpeed - rearAxleSpeed;
        float kp = Param::GetFloat(Param::tractionkp);

        //Here we assume front wheel drive
        if (diff < 0)
        {
            //Front axle turns slower than rear axle -> too much breaking force
            float speedErr = Param::GetFloat(Param::allowedlag) - diff;
            throtmin = -100 + (kp * speedErr);
        }
        else
        {
            //Front axle turns faster than rear axle -> wheel spin
            float speedErr = Param::GetFloat(Param::allowedspin) - diff;
            throtmax = 100 + (kp * speedErr);
        }
    }
}
```

Logical Operators: and / or / not

- “not a” $\rightarrow$ True if a is False
- “a and b” $\rightarrow$ True if a and b are True
- “a or b” $\rightarrow$ True if a or b (or both) are True
- Precedence: Order of execution. From First to Last:
 - not
 - and
 - or
- “a and b or not c” $\rightarrow$ “a and b or (not c)” $\rightarrow$ “(a and b) or (not c)”

else - Statement



else - Statement

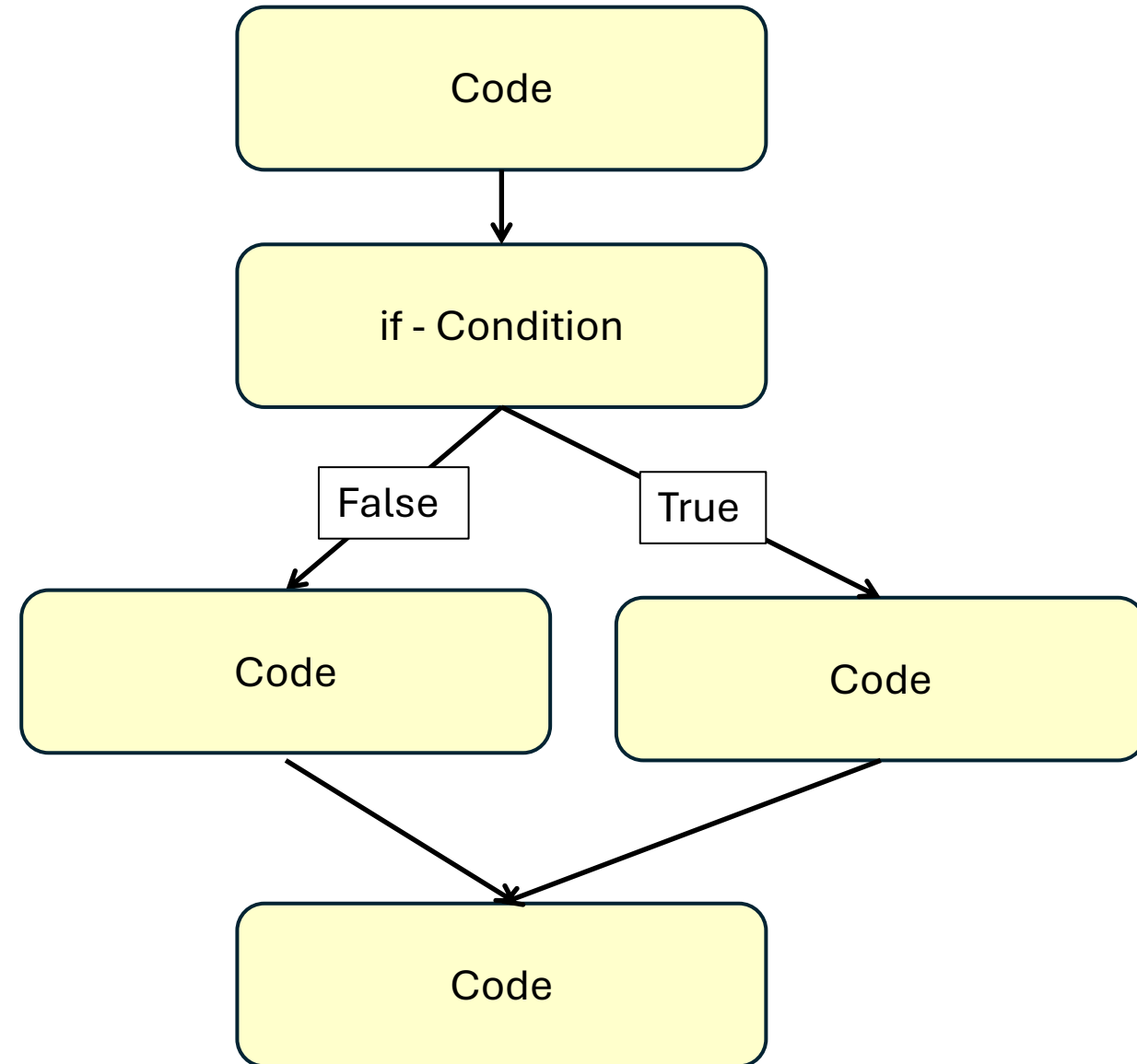
```
name = "Dieter"  
if name == "Emil":  
    print("The name is Emil")  
if name != "Emil":  
    print("The name is not Emil")
```

The name is not Emil



```
name = "Dieter"  
if name == "Emil":  
    print("The name is Emil")  
else:  
    print("The name is not Emil")
```

The name is not Emil



elif - Statement

- Not super important for now

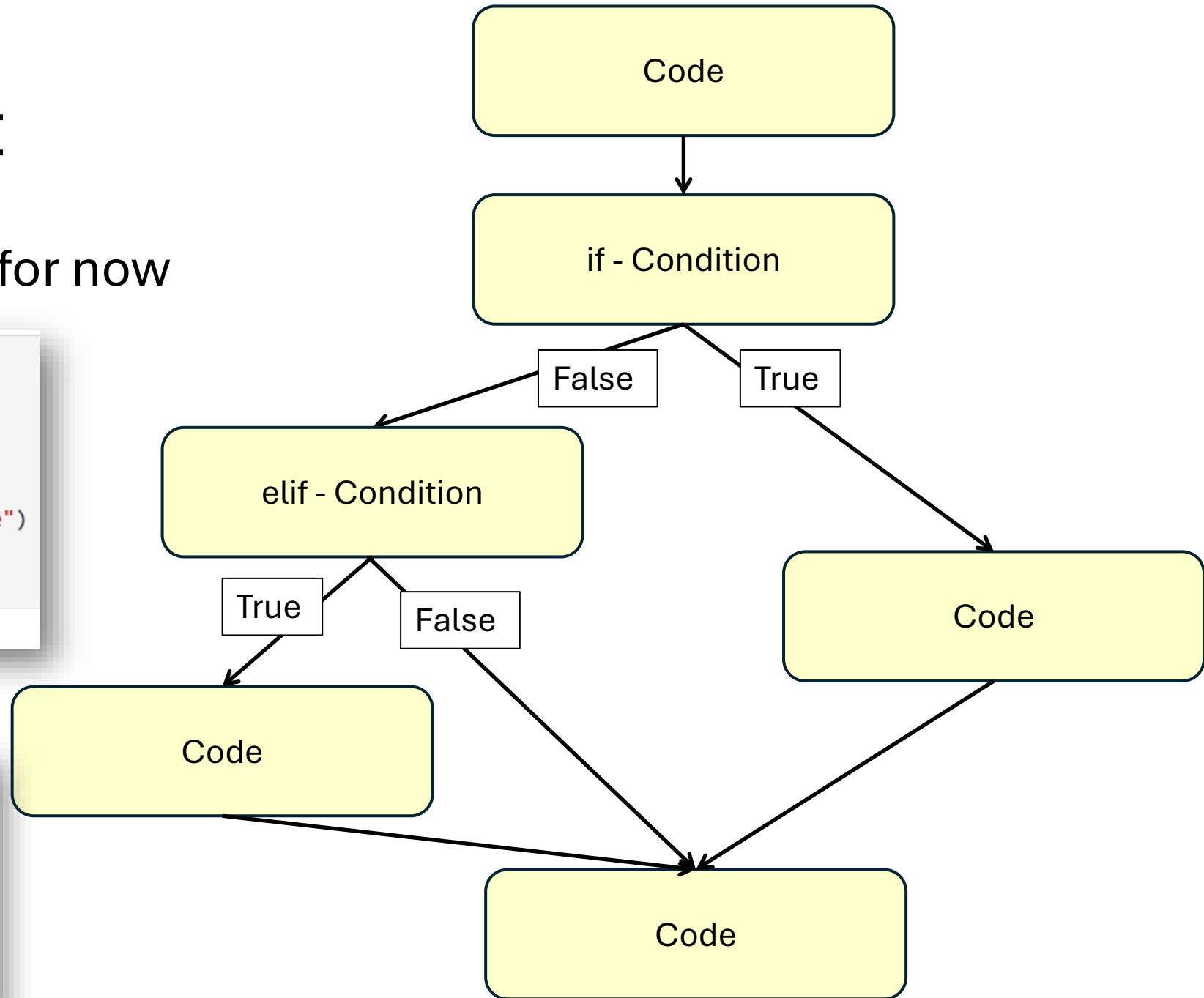
```
your_age = 12
if your_age < 16:
    print("You cannot buy alcohol")
else:
    if your_age >= 16 and your_age < 18:
        print("You can only buy beer and wine")
    else:
        print("You can buy all alcohol")
```

You cannot buy alcohol



```
your_age = 12
if your_age < 16:
    print("You cannot buy alcohol")
elif your_age >= 16 and your_age < 18:
    print("You can only buy beer and wine")
else:
    print("You can buy all alcohol")
```

You cannot buy alcohol



Extra: User Input

- “input” let’s the program ask you for an input:
→ Generates a string!

```
your_age = int(input("What is your age?"))
if your_age < 16:
    print("You cannot buy alcohol")
else:
    if your_age >= 16 and your_age < 18:
        print("You can only buy beer and wine")
    else:
        print("You can buy all alcohol")
```

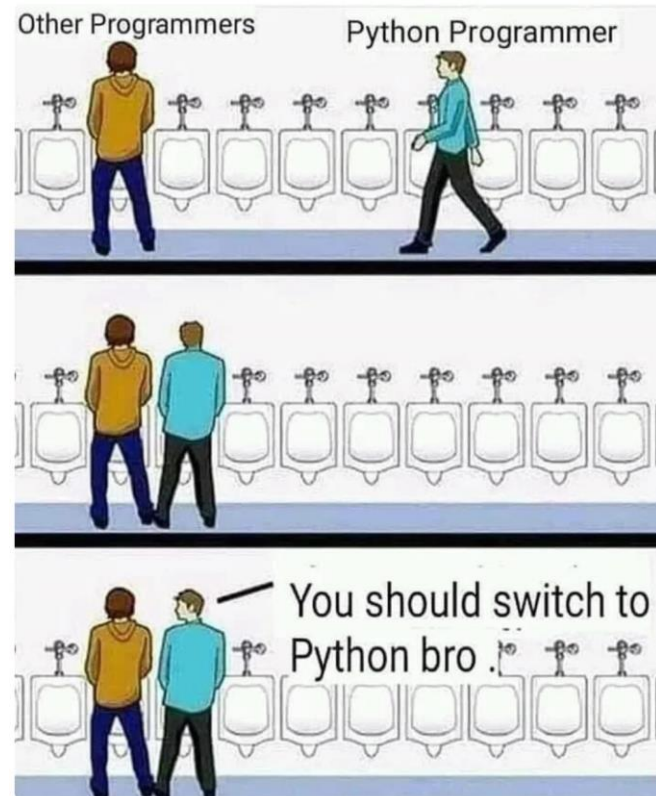
What is your age? ↑↓ for history. Search history with c-↑



```
your_age = int(input("What is your age?"))
if your_age < 16:
    print("You cannot buy alcohol")
else:
    if your_age >= 16 and your_age < 18:
        print("You can only buy beer and wine")
    else:
        print("You can buy all alcohol")
```

What is your age? 15
You cannot buy alcohol

Good Luck with your Tasks 😊



[<https://dev.to/codeitbro/15-funniest-and-hilarious-python-coding-memes-2aij>]